Hossein Khodavirdi

ACOUSTICS | WAVE MECHANICS | STRUCTURAL DYNAMICS | COMPUTATIONAL MECHANICS

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Professional Profile

Acoustics and mechanical-wave engineer whose work spans theoretical wave mechanics, computational modeling, and experimental characterization. Current work focuses on high-power piezoelectric transducers and acoustic architectures for semiconductor wafer processing. Research background includes acoustic and elastic wave scattering, open systems, mechanical metamaterials, photoacoustics, and physics-informed scientific computing. Experienced in connecting analytical predictions and multiphysics simulations with laboratory measurements and device-level design decisions.

Research Interests

Acoustic and elastic wave propagation; structural dynamics and vibration; wave scattering and open systems; mechanical and acoustic metamaterials; phononics and periodic media; piezoelectric transducers and high-power ultrasonics; computational mechanics and finite-element methods; Green's-function methods; physics-informed scientific computing; experimental vibration characterization; photoacoustics and ultrasound.

Research and Professional Appointments

Senior Acoustic Engineer | Crystal Sonic Inc.

July 2025–Present | Phoenix, Arizona

- Lead acoustic and transducer engineering for Crystal Sonic's Sonic Lift-Off wafer-separation technology, developing a novel high-power piezoelectric transducer and acoustic architecture for controlled elastic-wave energy delivery in semiconductor wafer processing under a DARPA-supported R&D program.
- Design and evaluate the transducer as a complete electromechanical system, combining piezoelectric elements, coupling layers, structural components, drive conditions, and wafer loading. Use coupled-field ANSYS modal, harmonic, transient, thermal, and structural analyses to assess resonance, acoustic loading, damping, prestress, lateral-thickness coupling, spurious modes, and stress localization.
- Plan and perform model-to-test validation using Polytec laser Doppler vibrometry, Bode 100 impedance spectroscopy, oscilloscopes, waveform generators, and RF power amplification. Analyze electrical impedance and spatial vibration measurements to identify modes, quantify response, calibrate damping and coupling assumptions, and resolve simulation-to-experiment discrepancies.
- Develop Python lumped-element and Butterworth–Van Dyke models of complete transducer stacks; fit measured impedance data and connect model parameters to device behavior. Design and assess LC matching networks and electrical drive conditions to improve power transfer over the target operating band.

- Build GPU-accelerated transient-wave and k-Wave workflows and validate predicted velocity and stress fields against spatial measurements. Investigate multi-element excitation, beamforming, and energy-delivery strategies to guide transducer architecture and system development.
- Investigate acoustic matching layers, engineered interfaces, temperature-dependent dynamics, high-drive piezoelectric behavior, and fracture-relevant stress fields in multilayer stacks; translate findings into design recommendations and experimental priorities.

Research Specialist | University of Illinois Chicago

August 2024–May 2025 | Chicago, Illinois

- Conducted computational and experimental research in photoacoustics and ultrasound.
- Developed Python and MATLAB workflows for multispectral photoacoustic microscopy, spectral unmixing, waveform decomposition, and quantitative image analysis.
- Implemented and compared FFT, least-squares, cross-correlation, and learning-based signal-decomposition methods; accelerated large waveform-processing workloads using multiprocessing and GPU computing.
- Modeled nonlinear thermoacoustic and photoacoustic signal generation under long-pulse excitation and compared physics-based predictions with Verasonics ultrasound measurements.
- Contributed to peer-reviewed Journal of Biophotonics publications.

Graduate Research Assistant | Wave Research Laboratory, Illinois Institute of Technology

January 2020–May 2025 | Chicago, Illinois | Advisor: Professor Ankit Srivastava

- Developed analytical and computational methods for acoustic and elastic wave scattering, mechanical metamaterials, periodic media, heterogeneous systems, and open-system wave propagation.
- Established causality- and passivity-based restrictions on acoustic and elastic constitutive properties using complex analysis and Herglotz–Nevanlinna theory.
- Developed an open-system formulation connecting resonant scattering to finite-dimensional effective non-Hermitian operators and finite-scatterer dynamics.
- Adapted the Atomistic Green's Function method to classical acoustic and elastic scattering, including finite phononic crystals and defects in two-dimensional elastic waveguides.
- Studied dynamic homogenization, local metamaterials, equi-frequency contours, Willis-type constitutive behavior, transition layers, and finite-sample scattering using plane-wave expansion, mode matching, COMSOL, FEniCS, Python, and MATLAB.
- Implemented JAX/Python physics-informed neural-network solvers for acoustic scattering with exact Dirichlet-to-Neumann radiation boundary conditions.
- Modeled acoustic scattering from multilayer elastic spherical shells using shell theory and spherical-harmonic expansions.
- Analyzed acoustic radiation force and torque on passive and internally actuated scatterers, including piezoelectrically driven acoustic steering.

Undergraduate Research Assistant | Iran University of Science and Technology

September 2014–December 2019 | Tehran, Iran

- Conducted research on acoustic radiation force and torque, acoustic steering of active spherical carriers, and vibration-based condition monitoring.

Education

Doctor of Philosophy in Mechanical Engineering | Illinois Institute of Technology

January 2020–May 2025 | Chicago, Illinois | GPA: 3.92/4.00

Thesis: Multiscale Approaches to Acoustic and Elastic Wave Scattering in Mechanical Metamaterials.

M.S. in Mechanical Engineering | Illinois Institute of Technology

2022 | Chicago, Illinois | GPA: 4.00/4.00

B.S. in Mechanical Engineering | Iran University of Science and Technology

2019 | Tehran, Iran | GPA: 3.48/4.00; senior-year GPA: 3.65/4.00

Undergraduate research: Acoustic Steering of Active Spherical Carriers.

Selected Research Contributions

- Open-system and Green's-function wave mechanics: non-Hermitian and Atomistic Green's Function frameworks for classical acoustic and elastic scattering, finite scatterers, defects, waveguides, and phononic systems.
- Passivity, causality, and material response: dispersion relations and frequency-domain restrictions for acoustic and elastic constitutive properties derived from causality, passivity, and complex analysis.
- Metamaterials and interfaces: local and nonlocal response, dynamic homogenization, equi-frequency surfaces, Willis coupling, transition layers, and finite-sample scattering.
- Piezoelectric transducers and high-power acoustics: measurement-calibrated multiphysics and reduced-order models incorporating electromechanical coupling, mode interaction, damping, acoustic loading, matching networks, and spatial energy delivery.
- Acoustic manipulation: analytical models of radiation force and torque and sustained pulling or steering of active spherical scatterers.
- Scientific computing: finite elements, plane-wave expansion, mode matching, Green's functions, physics-informed neural networks, GPU computing, and transient-wave solvers.

Journal Articles and Conference Proceedings

1. K. Yadollahpour, H. Khodavirdi, and A. Srivastava, "Bulk Locality and Interface Corrections in Acoustic Metamaterials," Journal of Sound and Vibration, accepted for publication, 2026. JSV.
2. A. Khansari, H. Khodavirdi, S. M. Ranjbaran, M. Zafar, M. J. Beirami, and K. Avanaki, "A Comparative Study on Signal Decomposition Techniques for Stimulated Raman Photoacoustic Microscopy," Journal of Biophotonics, 19(3), e70184, 2026.
3. A. Khansari, S. M. Ranjbaran, M. Zafar, N. Ghaffar Nia, H. Khodavirdi, M. Tsoukas, and K. Avanaki, "Multispectral Laser-Scanning Photoacoustic Microscopy With SRS-Generated Wavelengths for Skin Chromophore Characterization," Journal of Biophotonics, 19, e70225, 2026.
4. H. Khodavirdi, Z.-Y. Ong, and A. Srivastava, "The Atomistic Green's Function Method for Acoustic and Elastic Wave-Scattering Problems," International Journal of Mechanical Sciences, 275, 109263, 2024.
5. H. Khodavirdi and M. Rajabi, "Sustainable Pulling Motion of an Active Scatterer," Wave Motion, 119, 103134, 2023.

6. H. Khodavirdi, A. A. Mokhtari, and A. Srivastava, “Scattering of Mechanical Waves from the Perspective of Open Systems,” Mechanics of Materials, 172, 104399, 2022.
7. H. Khodavirdi and A. Srivastava, “The Analytical Structure of Acoustic and Elastic Material Properties,” Wave Motion, 108, 102837, 2022.
8. M. Rajabi, H. Khodavirdi, and A. Mojahed, “Acoustic Steering of Active Spherical Carriers,” Ultrasonics, 105, 106112, 2020.
9. A. Khansari, S. M. Ranjbaran, M. Zafar, N. Ghaffar Nia, H. Khodavirdi, M. Tsoukas, and K. Avanaki, “Spectral unmixing of skin chromophores using SRS and regularized NNLS,” SPIE Photonics in Dermatology and Plastic Surgery, 2026.

Manuscripts in Preparation

- H. Khodavirdi, M. J. Politis, N. Sukumar, and A. Srivastava, “Exact Imposition of Dirichlet-to-Neumann Maps in Physics-Informed Neural Networks for an Acoustic Wave-Scattering Problem,” manuscript in preparation.

Selected Conference Presentations

- 2023 — “Exact Imposition of DtN Maps in Physics-Informed Neural Networks for an Acoustic Wave-Scattering Problem,” 17th U.S. National Congress on Computational Mechanics, New Mexico. With M. J. Politis, N. Sukumar, and A. Srivastava.
- 2022 — “The Atomistic Green’s Function Method for Acoustic and Elastic Wave-Scattering Problems,” ASME International Mechanical Engineering Congress and Exposition, Columbus, Ohio. With Z.-Y. Ong and A. Srivastava.
- 2022 — “Scattering of Mechanical Waves from the Perspective of Open Systems,” U.S. National Congress of Theoretical and Applied Mechanics, Austin, Texas. With A. A. Mokhtari and A. Srivastava.
- 2021 — “The Analytical Structure of Acoustic and Elastic Material Properties,” ASME International Mechanical Engineering Congress and Exposition. With A. Srivastava.

Teaching Experience

Adjunct Faculty, MMAE 350 Computational Mechanics | Illinois Institute of Technology

Spring 2024 | Chicago, Illinois

- Taught undergraduate computational mechanics and numerical methods; developed lectures, assignments, examinations, computational examples, and student assessments.
- Connected numerical linear algebra, differential equations, finite-difference and finite-element methods, and engineering computation to mechanics and simulation.
- Emphasized physical interpretation, verification, and responsible use of computational tools.

Additional teaching experience: supported undergraduate mechanical-engineering instruction through grading, technical feedback, and student assistance (Spring 2025).

Courses prepared to teach: Engineering Dynamics; Mechanical Vibrations; Computational Mechanics; Numerical Methods; Finite Element Analysis; Structural Dynamics; Acoustics; Wave Propagation; Mechanical Metamaterials and Phononics; Piezoelectric Transducers and Ultrasonics.

Sponsored Research Experience

- DARPA-supported SBIR Phase II program, Crystal Sonic Inc. (2025–present): transducer design, computational modeling, experimental characterization, and acoustic-energy-delivery research for Sonic Lift-Off semiconductor layer-separation technology.
- NSF-supported research, Illinois Institute of Technology (2022–2025): theoretical and computational work on acoustic and elastic metamaterials, boundaries and interfaces, transition layers, and finite-sample scattering.

Honors and Awards

- Outstanding Research Assistant Award, Department of Mechanical, Materials, and Aerospace Engineering, Illinois Institute of Technology, 2024.
- Conference Award, 17th U.S. National Congress on Computational Mechanics, 2023.

Professional Service and Leadership

- Peer reviewer for Finite Elements in Analysis and Design, Mechanics of Materials, and Wave Motion.
- Vice President, Graduate Student Advisory Council, Department of Mechanical, Materials, and Aerospace Engineering, Illinois Institute of Technology, 2022.

Technical Expertise

Acoustics and wave mechanics: Acoustic and elastic wave propagation, scattering, ultrasonics, vibration, resonators, phononics, periodic media, metamaterials, radiation force and torque, beamforming, acoustic impedance, and power transfer.

Piezoelectric and ultrasonic systems: Piezoelectric transducers, high-power ultrasonics, electromechanical coupling, thickness and lateral modes, multi-element arrays, damping, quality factor, acoustic loading, matching layers, impedance matching.

Experimental methods: Polytec LDV, Bode 100 impedance analysis, Verasonics ultrasound acquisition, oscilloscopes, waveform generators, RF power amplifiers, vibration and resonance characterization.

Computational mechanics: ANSYS coupled-field, modal, harmonic, transient, and thermal-structural analysis; COMSOL; FEniCS/FEniCSx; k-Wave; finite elements and differences; eigenvalue and waveguide problems.

Analytical and numerical methods: Green's functions, Atomistic Green's Function method, plane-wave expansion, mode matching, spherical harmonics, variational and spectral methods, Dirichlet-to-Neumann maps, lumped-parameter and Butterworth–Van Dyke models.

Scientific computing and design: Python, MATLAB, C++, JAX, NumPy/SciPy, PyTorch, TensorFlow/Keras, GPU/CUDA workflows, LTspice, Qucs, and SolidWorks.

Selected Graduate Coursework

Wave Propagation; Advanced Dynamics; Advanced Mechanics of Solids; Continuum Mechanics; Advanced Engineering Mathematics I and II; Applied Analysis; Optimization for Engineers; Data-Driven Modeling; Deep Learning; Quantum Computing.